

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1753

South Akron 100 kV
April 21, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1753. The Interconnection Customer proposes a 182.4 MW solar photovoltaic facility interconnecting at the South Akron 100 kV substation in the Ohio area. The request was received on May 19, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 429 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$12,429,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: TALLMADGE to EVANS 100 KV
Contingency: TALLMADGE to EVANS 100 KV Circuit 2
Loading: 121.2% (Rating: 157 MVA)
DFAX: 12.7%
Pre-project loading: 106.4%
2. Facility: EDGERTON to EVANS SUBSTATION 100 KV
Contingency: EDGERTON to EVANS 100 KV Ckt 1
Loading: 118.6% (Rating: 179 MVA)
DFAX: 11.9%
Pre-project loading: 106.4%
3. Facility: SOUTH AKRON to MOORE 100 KV Ckt 2
Contingency: MOORE to SOUTH AKRON 100 KV Cir. 1
Loading: 114.0% (Rating: 169 MVA)
DFAX: 26.7%
Pre-project loading: 85.2%
4. Facility: SOUTH AKRON to MOORE 100 KV Ckt 1
Contingency: MOORE to S KRON 100 KV Ckt 2
Loading: 112.2% (Rating: 171 MVA)
DFAX: 26.7%
Pre-project loading: 83.8%
5. Facility: EVANS to EDGERTON 100 KV Cir. 1
Contingency: EDGERTON to EVANS 100 KV Cir. 2
Loading: 107.6% (Rating: 197 MVA)
DFAX: 11.9%
Pre-project loading: 96.6%

6. Facility: EVANS to TALLMADGE 100 KV line
Contingency: EVANS to TALLMADGE 100 KV
Loading: 105.6% (Rating: 180 MVA)
DFAX: 12.7%
Pre-project loading: 92.7%
7. Facility: EDGERON to WATER 100 KV Circuit 1
Contingency: WATER to EDGERTON 100 KV Line 2
Loading: 101.1% (Rating: 214 MVA)
DFAX: 8.5%
Pre-project loading: 93.9%
8. Facility: BELL ROCK ROAD to MOORE 100 KV
Contingency: PORTAGE LAKES 100-1 kV Transformer
Loading: 101.0% (Rating: 179 MVA)
DFAX: 21.0%
Pre-project loading: 79.5%
9. Facility: TALLMDGE SUBSTATION to DAVID SWINFORD 100 KV L
Contingency: OLD FORGE to PORTAGE LAKES 230 KV line
Loading: 101.0% (Rating: 206 MVA)
DFAX: 12.1%
Pre-project loading: 90.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
TALLMADGE to EVANS 100 KV Cir. 1	Rebuild line with higher-capacity conductor	\$1,068,000	100% (\$1,068,000)
EDGERTON to EVANS SUBSTATION 100 KV	Rebuild line with higher-capacity conductor	\$2,834,000	100% (\$2,834,000)
MOORE to S AKRON 100 KV Circuit 2	Rebuild line with higher-capacity conductor	\$1,793,000	100% (\$1,793,000)
SOUTH AKRON to MOORE 100 KV Ckt 1	Rebuild line with higher-capacity conductor	\$3,075,000	100% (\$3,075,000)
EVANS to EDGERTON 100 KV Cir. 1	Reconductor line	\$902,000	100% (\$902,000)
TALLMADGE to EVANS 100 KV Cir. 2	Reconductor line	\$566,000	100% (\$566,000)
EDGERON to WATER 100 KV Circuit 1	Reconductor line	\$730,000	100% (\$730,000)
BELL ROCK ROAD to MOORE 100 KV	Reconductor line	\$1,394,000	76% (\$1,059,000)
TALLMDGE SUBSTATION to DAVID SWINFORD 100 KV L	Reconductor line	\$402,000	100% (\$402,000)
	Total allocated to project		\$12,429,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.